Maths Explorer

Progressive exercises

E1 · Direct application. Determine all antiderivatives on $\mathbb{R}$ of the function $x \mapsto 3x^2 - 4x + 1$.

E2 · Application and justification. Determine the antiderivative F of the function $x \mapsto \frac{2x}{x^2+4}$ on $\mathbb{R}$ who verifies $F(0) = 0$.

E3 · Method to be organised. Determine the solution on $\mathbb{R}$ differential equation $y' = 3y$ who checks $y(1) = 2$.

E4 · Choice of strategy. Determine the solution on $\mathbb{R}$ differential equation $y' = -y + 4$ which checks $y(0) = 7$.

E5 · Synthesis or demonstration. The differential equation is considered $(E) : y' = y - x$ on $\mathbb{R}$. Verify that $g(x) = x + 1$ is a solution, then determine the solution of (E) who verifies $y(0) = 0$.